KAIZEN MATHS

Lesson starter

Ratio and Proportion

Ratio Applications

Name

S1

On a map with scale 1:5000, the distance between two towns is 6 cm. Find the real distance.

S2

A mixture uses cement and aggregate in the ratio 2:2. How much of each is needed for 20tonnes?

S3

The exchange rate is £1 = $1.42. Convert £512 to US dollars.

S4

On a map with scale 1:1000, the distance between two towns is 9 cm. Find the real distance.

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NUM-B9A2-L2 · S

Show your working

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Exit ticket

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A recipe uses 274g flour and 119g sugar. How much of each is needed to make 4 times the recipe?

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Teacher answers & working

Ratio and Proportion · Ratio Applications

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S1

On a map with scale 1:5000, the distance between two towns is 6 cm. Find the real distance.

Answer:

300 m

Working:

$1\text{ cm} : 5000\text{ cm} = 6\text{ cm} : ?\text{ cm}$

Real distance = 6 × 5000 = 30000 cm

Convert centimetres to metres by dividing by 100.

$30000\text{ cm} = 300\text{ m}$

A scale of 1:5000 means 1 cm on the map represents 5000 cm in reality.

S2

A mixture uses cement and aggregate in the ratio 2:2. How much of each is needed for 20tonnes?

Answer:

10tonnes cement, 10tonnes aggregate

Working:

Total parts = 2 + 2 = 4.

One part = 20tonnes ÷ 4 = 5tonnes

cement = 2 × 5 = 10tonnes

aggregate = 2 × 5 = 10tonnes

Check: 10tonnes + 10tonnes = 20tonnes

Construction mixtures depend on proportional material quantities.

S3

The exchange rate is £1 = $1.42. Convert £512 to US dollars.

Answer:

$727.04

Working:

This is direct proportion: more pounds convert to more dollars.

$£1 : \mathbf{1.42} = \mathbf{£512} :?$

Multiply the amount in pounds by the dollar value of £1.

$512 \times 1.42 = 727.04$

Multiply by the exchange rate to convert pounds to dollars.

Teacher answers & working

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S4

On a map with scale 1:1000, the distance between two towns is 9 cm. Find the real distance.

Answer:

90 m

Working:

$1\text{ cm} : 1000\text{ cm} = 9\text{ cm} : ?\text{ cm}$

Real distance = 9 × 1000 = 9000 cm

Convert centimetres to metres by dividing by 100.

$9000\text{ cm} = 90\text{ m}$

A scale of 1:1000 means 1 cm on the map represents 1000 cm in reality.

E1

A recipe uses 274g flour and 119g sugar. How much of each is needed to make 4 times the recipe?

Answer:

1096g flour, 476g sugar

Working:

Scale every ingredient by the same multiplier.

Flour = 274 × 4 = 1096g

Sugar = 119 × 4 = 476g

Recipe scaling keeps all ingredient proportions the same.

E2

The exchange rate is £1 = $1.22. Convert £414 to US dollars.

Answer:

$505.08

Working:

This is direct proportion: more pounds convert to more dollars.

$£1 : 1.22 = £414 : ?$

Multiply the amount in pounds by the dollar value of £1.

$414 \times 1.22 = 505.08$

Multiply by the exchange rate to convert pounds to dollars.

Teacher answers & working

Ratio and Proportion · Ratio Applications

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E3

On a map with scale 1:2500, the distance between two towns is 9 cm. Find the real distance.

Answer:

225 m

Working:

1 cm : 2500 cm = 9 cm : ? cm

Real distance = 9 × 2500 = 22500 cm

Convert centimetres to metres by dividing by 100.

22500 cm = 225 m

A scale of 1:2500 means 1 cm on the map represents 2500 cm in reality.